Vishakha Ramani

Research Staff Member, IBM Thomas J. Watson Research Center
Yorktown Heights, New York, USA

vishakharamani@ibm.com — Google Scholar — LinkedIn

Research Interests

Performance modeling and analysis of large-scale systems; queueing theory and stochastic optimization; LLM inference serving; Age of Information; cloud-native infrastructure for AI workloads.

Education

Rutgers University–New Brunswick 2020–2024
Ph.D., Electrical and Computer Engineering
Dissertation: *Storing, Retrieving, and Processing Updates: A Timeliness Perspective*
Advisor: Prof. Roy D. Yates (WINLAB)

Rutgers University 2017–2019
M.S., Electrical and Computer Engineering
Thesis: *I-Mac: An ICN Based Radio Access Network Architecture*
Advisor: Prof. Dipankar Raychaudhuri

Rajiv Gandhi Prodyogiki Vishwavidyalaya 2011–2016
B.Tech. and M.Tech. (Dual Degree), Electronics and Communication Engineering

Professional Experience

Research Staff Member October 2024–Present
IBM Thomas J. Watson Research Center, Yorktown Heights, NY

- Co-architect of the Workload Variant Autoscaler (WVA) for *llm-d*, an open-source Kubernetes-native LLM inference system (GitHub; architecture docs). Produced the first end-to-end system design and led the autoscaling effort with the broader llm-d community.
- Designed and implemented the queueing model analyzer at the analytical core of WVA: a hardware-aware performance model of LLM inference that predicts time-to-first-token and inter-token latency, and computes the maximum sustainable request rate per replica under SLO constraints. Model parameters are learned online via a Kalman filter, removing the need for offline profiling.
- Co-authored the WVA system paper, A. Malvankar et al., "WVA: A Global Optimization Control Plane for llm-d," IEEE CLOUD 2026 (to appear; arXiv:2603.09730).
- Contributed analysis to T. Abdelzaher et al., "The Bottlenecks of AI," *Real-Time Systems*, vol. 61, 2025: framing open problems in scaling inference serving under latency SLOs, GPU memory management, and orchestration of heterogeneous accelerators.
- Developing an analytical framework for prefill/decode (P/D) disaggregation, deriving a dynamic scheduling threshold via Lyapunov optimization on a tandem queueing model with provable throughput and latency guarantees (manuscript in preparation).
- Applying the AI-Driven Research for Systems (ADRS) methodology of Liu et al. (arXiv:2510.06189) to scheduling and autoscaling problems in llm-d, contributing analytical performance models that supply provable guarantees for algorithms discovered by the agentic search loop. Related open-source team artifacts: BLIS (simulation substrate) and Agentic Strategy Evolution (search loop).

Graduate Research Assistant September 2018–October 2024
Rutgers University–New Brunswick
Ph.D. research — Age of Information (AoI) in computing systems

- **AoI in shared memory and synchronization.** Analyzed the impact of synchronization primitive choice (lock-based reader-writer locks vs. lock-free Read-Copy-Update) on information freshness in shared-memory status-updating systems. Showed that the optimal choice depends on the update rate regime: lock-based primitives favor high-rate, lock-free favors low-rate workloads. Characterized age–memory trade-offs in RCU systems (INFOCOM 2023; INFOCOM 2024 Workshop).
- **Optimal memory access policies.** Formulated an optimal memory sampling problem in status-updating systems and proved the optimal policy is a stationary, deterministic threshold-type policy; derived closed-form AoI expressions (ISIT 2024).
- **Multi-source and multi-step update processing.** Developed analytical models for AoI in publish–subscribe systems with multiple sources, showing a lazy computation policy can reduce average AoI at the monitor (WiOpt 2023). Extended to multi-step pipelines, identifying wasted computation when processing effort does not reduce age, and optimized service rates under power constraints (Asilomar 2024).
- **Experimental timeliness in mobile networks.** DPDK-based testbed studies of timely packet routing, demonstrating AoI sensitivity to offered load and concurrency constructs in real deployments (INFOCOM 2023 Workshop).

M.S. research — ICN-based wireless access networks

- Designed an Information-Centric Networking (ICN) based RAN protocol for LTE enabling efficient wireless multicast via a cross-layer architecture integrating ICN identifiers with LTE RAN; verified with NS-3 and SUMO simulations.
- Designed a scalable ICN-based MAC scheduling algorithm using convex optimization for joint resource allocation across unicast and multicast user groups.

Research Intern May–August 2023
IBM Thomas J. Watson Research Center

- Performance analysis of the Multi-Cluster App Dispatcher (MCAD) using Kubernetes Without Kubelet (KWOK) to identify scheduling bottlenecks in cloud-native foundation model workloads.
- Presented findings at KubeCon + CloudNativeCon North America 2023.

Research Intern May–August 2022
IBM Thomas J. Watson Research Center

- Built a middleware stack for distributed training of foundation models on OpenShift as part of Project Codeflare.
- Contributed open-source code to TorchX.
- Benchmarked multi-tenant cluster schedulers under concurrent foundation model training workloads.

Assistant System Engineer February–July 2017
Tata Consultancy Services, Mumbai, India

- Business intelligence and performance management; developed BI solutions using IBM Cognos.

Publications

Journal Articles

1. T. Abdelzaher, Y. Hu, D. Kara, T. Kimura, A. Misra, **V. Ramani**, O. Tardieu, et al., "The Bottlenecks of AI: Challenges for Embedded and Real-Time Research in a Data-Centric Age," *Real-Time Systems*, vol. 61, no. 2, pp. 185–236, 2025.
2. **V. Ramani** and S. K. Sharma, "Cognitive Radios: A Survey on Spectrum Sensing, Security and Spectrum Handoff," *China Communications*, vol. 14, no. 11, pp. 185–208, 2017.

Conference Papers

1. A. Malvankar, L. Villard, M. Abdi, E. Shindin, B. Dumba, **V. Ramani**, A. Tantawi, et al., "WVA: A Global Optimization Control Plane for llm-d," *Proc. IEEE International Conference on Cloud Computing (CLOUD)*, 2026 (to appear; arXiv:2603.09730).
2. **V. Ramani**, I. Seskar, and R. D. Yates, "Timely and Energy-Efficient Multi-Step Update Processing," *Proc. 58th Asilomar Conference on Signals, Systems, and Computers*, pp. 116–120, 2024.
3. **V. Ramani**, I. Seskar, and R. D. Yates, "Efficient and Timely Memory Access," *Proc. IEEE International Symposium on Information Theory (ISIT)*, pp. 1155–1160, 2024.
4. **V. Ramani**, I. Seskar, and R. D. Yates, "Timely Processing of Updates from Multiple Sources," *Proc. 21st International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, 2023.
5. **V. Ramani**, J. Chen, and R. D. Yates, "Lock-Based or Lock-Less: Which is Fresh?" *Proc. IEEE INFOCOM*, pp. 1–10, 2023.

Workshop Papers

1. **V. Ramani**, J. Chen, and R. D. Yates, "Age-Memory Trade-Off in Read-Copy-Update," *IEEE INFOCOM Workshops*, 2024.
2. **V. Ramani**, J. Chen, and R. D. Yates, "Timely Mobile Routing: An Experimental Study," *IEEE INFOCOM Workshops*, 2023.

Manuscripts in Preparation

1. **V. Ramani** and A. N. Tantawi, "An Approximate Parameterized Queueing Model of LLM Inference Serving." Tractable Markovian queueing model; closed-form mean ITL and numerical evaluation of TTFT; validated against GPU measurements.
2. **V. Ramani**, "When to Disaggregate: Optimal Online Scheduling for Prefill-Decode Separated LLM Inference."

Theses

1. **V. Ramani**, "Storing, Retrieving, and Processing Updates: A Timeliness Perspective," Ph.D. Dissertation, Rutgers University, 2024.
2. **V. Ramani**, "I-Mac: An ICN Based Radio Access Network Architecture," M.S. Thesis, Rutgers University, 2020.

Talks and Presentations

- "Performance Analysis of Multi-Cluster App Dispatcher (MCAD)," KubeCon + CloudNativeCon North America 2023, Chicago, IL, November 2023.
- "Using Age of Information (AoI) Models to Enable Low Latency Services," Fall 2022 Research Review, WINLAB, Rutgers University.

Professional Service

Program Committees

- AAAI/ACM Conference on AI, Ethics, and Society (AIES) 2026, Malmö, Sweden.

Reviewer for

- *IEEE Journal on Selected Areas in Communications* (JSAC)
- *IEEE Transactions on Wireless Communications* (TWC)
- *IEEE Transactions on Communications* (TCOM)
- *IEEE International Symposium on Information Theory* (ISIT)
- *IEEE Internet of Things Journal*
- *Ad Hoc Networks* (Elsevier)
- *BalkanCom*

Invention Disclosures

- A. Malvankar, A. Tantawi, **V. Ramani**, T. Eilam, M. Abdi, "System and Method for Efficient SLO-Aware Autoscaling of AI Inference Workloads," IBM Invention Disclosure P202503087, 2025.

Honors and Awards

- CNCF Travel Grant, KubeCon + CloudNativeCon North America 2023.
- Graduate Research Assistantship, Rutgers University (WINLAB), 2018–2024.
- Teaching Assistantship, Rutgers University, ECE Department.

Technical Skills

Methods: Queueing theory, Lyapunov optimization, stochastic processes, convex optimization, Age of Information analysis
Systems: Kubernetes, OpenShift, DPDK, NS-3, SUMO
Languages: Python, Go, C, C++, MATLAB, SQL, Verilog